9th Bangladesh Olympiad on Astronomy and Astrophysics

National Round 2026 – Senior

Date

Instructions for the Candidate:

- For all questions, the process involved in arriving at the solution is more important than the answer itself. Valid assumptions / approximations are perfectly acceptable. Please write your method clearly, explicitly stating all reasoning.
প্রতিটি প্রশ্নের জন্যই উত্তরের চেয়ে সমাধানের প্রক্রিয়া বেশি গুরুত্বপূর্ণ। যুক্তিপূর্ণ অনুমান/অ্যাপ্রক্সিমেশন পুরোপুরিভাবে গ্রহণযোগ্য। সমাধানের বিশদ ও স্পষ্ট ব্যাখ্যা আমাদের প্রত্যাশিত।
- Be sure to calculate the final answer in the appropriate units asked in the question.
চূড়ান্ত উত্তর প্রশ্ন অনুযায়ী সঠিক এককে গ্রহণযোগ্য।
- Non-programmable scientific calculators are allowed.
নন প্রোগ্রামেবল সায়েন্টিফিক ক্যালকুলেটর গ্রহণযোগ্য।
- The mark distribution is shown in the [ ] at the right corner for every question.
প্রতি প্রশ্নের শেষে [ ] বন্ধনীতে নম্বর বন্টন দেয়া আছে
- Time duration is *2 hours*
সময় ২ ঘন্টা

নাম(বাংলায়) :
নাম (In English) :
শ্রেণি (২০২৬ সাল) : মোবাইল নং:
প্রতিষ্ঠান (Institution) :
জন্মতারিখ (dd/mm/yy) :

Useful Constants and Formulas

Mass of the Sun	$M_\odot$	$\approx$	$1.989 \times 10^{30}\,\text{kg}$
Radius of the Sun	$R_\odot$	$\approx$	$6.955 \times 10^{8}\,\text{m}$
Mass of the Earth	$M_\oplus$	$\approx$	$5.972 \times 10^{24}\,\text{kg}$
Radius of the Earth	$R_\oplus$	$\approx$	$6.371 \times 10^{6}\,\text{m}$
Mass of the Moon	$M_☾$	$\approx$	$7.347 \times 10^{22}\,\text{kg}$
Distance from the Earth to the Moon	$r_{☾-\oplus}$	$=$	384400 km
Radius of the Moon	$R_☾$	$=$	1737 km
Speed of light	c	$\approx$	3×10^{8} m/s
Synodic period of Moon rotation		$\approx$	29.5 days
Astronomical Unit (AU)	$a_\oplus$	$\approx$	$1.496 \times 10^{11}\,\text{m}$
Solar Luminosity	$L_\odot$	$\approx$	$3.826 \times 10^{26}\,\text{W}$
Solar Constant	$S_\odot$	$\approx$	$1367\,\text{W/m}^2$
Gravitational Constant	G	$\approx$	$6.674 \times 10^{-11}\,\text{Nm}^2\text{kg}^{-2}$
Stefan's constant	σ	$=$	$5.670 \times 10^{-8}\,\text{Wm}^{-2}\text{K}^{-4}$
Boltzmann's Constant	k_B	$=$	$1.38 \times 10^{-23}\,\text{JK}^{-1}$
Hubble Constant	H_0	$\approx$	70 km/s/Mpc

1 Occultations and the Aldebaran Cycle [17]

When the Moon passes in front of a distant star, the event is called an *occultation*. Because the Moon is close to the Earth, its apparent position depends on the observer's location, introducing a parallax effect. As a result, an occultation may be visible only from certain regions of the Earth. Consider a star located at fixed ecliptic latitude $\beta_\star$. The Moon moves in a circular orbit inclined by an angle i with respect to the ecliptic. The line of nodes of the Moon's orbit precesses uniformly with period T_{pr}.
Let:

- m denote the Moon's angular semi-diameter,
- p denote the Moon's horizontal parallax,
  Assuming:
$$m = \arcsin\left(\frac{R_☾}{\ell_☾}\right), \qquad p = \arcsin\left(\frac{R_\oplus}{\ell_☾}\right),$$
- d denote the geocentric angular separation between the Moon's centre and the star.

1. Using geometric reasoning, determine the **maximum value of** d for which an occultation of the star by the Moon is possible for *some observer on Earth*. Your derivation should account for the effect of parallax and must be expressed in terms of m and p. [2]
2. Assume that the Moon crosses the ecliptic longitude of the star. At that instant, its ecliptic latitude $\beta_☾$ varies over time due to nodal precession.
   At that instant, the angular separation may be approximated as $d \approx |\beta_\star - \beta_☾|$.
   (a) Using your result from part (a), determine the **maximum allowed magnitude of** $\beta_☾$ for which occultations of the star are still possible. [2]
   (b) Then, assuming that the Moon's latitude varies as
$$\beta_☾ = i \sin\lambda,$$
   where λ is the Moon's ecliptic longitude and increases uniformly with time over a full cycle, determine the total range of λ for which occultations occur. [4]

Rohini (Aldebaran, or α Tauri) is the brightest star that the Moon can occult. Rohini is located in the constellation of Taurus, and its name translates as the "Reddish One." This star holds particular importance in Indian astronomy. Its frequent close approaches to the Moon influenced ancient astronomical traditions, including the system of the 27 *Nakshatras* (lunar mansions), which were mythologically regarded as the daughters of Daksha Brahma and consorts of the Moon. Among them, Rohini was considered the Moon's favourite, partly because occultations of Rohini occur relatively often.

3. Aldebaran (α Tau) has ecliptic latitude $\beta_\star = -5.47^\circ$. The inclination of the Moon's orbit is $i = 5.15^\circ$, and the nodal precession period is $T_{\text{pr}} = 18.6$ years.
   (a) Evaluate the critical latitude condition numerically. [3]
   (b) Determine the total duration of the interval during which monthly occultations of Aldebaran occur. [4]
   (c) If such a cycle begins in February 2015, determine the month and year when it ends. [2]

Solutions

Solution a

The occultation condition follows from a simple geometric argument. From a geocentric viewpoint the Moon's disc spans an angular radius m. However, an observer on the surface of the Earth is displaced from the geocentre by up to one Earth radius, which causes the Moon's apparent direction to shift by at most the horizontal parallax p. Hence the Moon can cover the star for *some* terrestrial observer if and only if the geocentric separation satisfies

$$\boxed{d_{\max} = m + p.}$$

[Correct geometric argument **1 Mark**; correct expression **1 Mark**]

Solution b

When the Moon crosses the ecliptic longitude of the star the angular separation reduces to

$$d = |\beta_\star - \beta_☾|.$$

Substituting into the condition from part (a), occultation is possible when

$$|\beta_\star - \beta_☾| \leqslant m + p.$$

Rearranging, the allowed range of lunar ecliptic latitude is

$$\beta_\star - (m + p) \ \leqslant\ \beta_☾ \ \leqslant\ \beta_\star + (m + p).$$

The maximum allowed **magnitude** of $\beta_☾$ for which occultation is still geometrically possible is therefore

$$\boxed{|\beta_☾|_{\max} = |\beta_\star| + (m + p).}$$

(The exact one-sided bound depends on the sign of $\beta_\star$ and is evaluated numerically in part C.1.)

[Correct inequality setup **1 Mark**; correct expression for $|\beta_☾|_{\max}$ **1 Mark**]

Solution c

With $\beta_☾ = i \sin\lambda$, occultation occurs when

$$|\beta_\star - i\sin\lambda| \leqslant m + p,$$

i.e. when

$$\frac{\beta_\star - (m+p)}{i} \ \leqslant\ \sin\lambda \ \leqslant\ \frac{\beta_\star + (m+p)}{i}.$$

Since $|\beta_\star| < i$ (the star is reachable by the Moon), the upper bound exceeds -1 and the lower bound may be below -1; the Moon's maximum southern excursion $-i$ provides a physical floor. Define

$$\kappa = \arcsin\left(\frac{|\beta_\star| - (m+p)}{i}\right),$$

then in the half-cycle where $\sin\lambda < 0$ the occultation window spans

$$\lambda \in [180^\circ + \kappa,\ 360^\circ - \kappa],$$

giving a total range

$$\boxed{\gamma = 2(90^\circ - \kappa)\,.}$$

[Correct trigonometric inequality **2 Mark**; correct expression for γ **2 Mark**]

Solution d

Using $R_☾ = 1.7374 \times 10^6\,\mathrm{m}$, $R_\oplus = 6.371 \times 10^6\,\mathrm{m}$, and $\ell_☾ = 3.84 \times 10^8\,\mathrm{m}$:

$$m = \arcsin\left(\frac{1.7374 \times 10^6}{3.84 \times 10^8}\right) \approx 0.2596^\circ, \qquad p = \arcsin\left(\frac{6.371 \times 10^6}{3.84 \times 10^8}\right) \approx 0.9505^\circ,$$

$$m + p \approx 1.210^\circ.$$

[Correct numerical values of m and p **1 Mark**]

With $\beta_\star = -5.47^\circ$, the occultation condition $|\beta_\star - \beta_☾| \leqslant m + p$ becomes

$$-6.680^\circ \;\leqslant\; \beta_☾ \;\leqslant\; -4.260^\circ.$$

Since the Moon's latitude is bounded by $\pm i = \pm 5.15^\circ$, the lower bound -6.680° is not physically reached. The effective occultation window is

$$\boxed{-5.15^\circ \;\leqslant\; \beta_☾ \;\leqslant\; -4.260^\circ.}$$

[Correct numerical window for $\beta_☾$ **2 Mark**]

Solution e

Apply the result of part (b.2) with the Aldebaran numbers:

$$\kappa = \arcsin\left(\frac{5.47^\circ - 1.210^\circ}{5.15^\circ}\right) = \arcsin(0.8272) \approx 55.78^\circ.$$

The total occultation range per precession cycle is

$$\gamma = 2(90^\circ - 55.78^\circ) = 68.44^\circ.$$

Since λ sweeps 360° in one nodal precession period T_{pr}, the duration of the occultation interval is

$$\Delta t = T_{\mathrm{pr}}\,\frac{\gamma}{360^\circ} = 18.6 \times \frac{68.44^\circ}{360^\circ} \approx \boxed{3.53\ \text{years.}}$$

[Correct κ and γ **2 Mark**]; [Correct duration Δt **2 Mark**]

Solution f

Adding $\Delta t \approx 3.53$ years to February 2015 gives

$$\text{February } 2015 + 3.53\ \text{yr} \approx \text{August } 2018.$$

$$\boxed{\text{The cycle ends in August 2018.}}$$

[Correct month and year (± 1 month acceptable) **2 Mark**]

2 Olbers's Paradox [14]

In 1823, Heinrich Olbers posed a deceptively simple question: *why is the night sky dark?* But you might ask—why is this even a question? Let us find the source of this paradox.
We define the **angular area** (solid angle) of an object in the sky as $\Omega = \mathcal{A}/r^2$, where $\mathcal{A}$ is the cross-sectional area of the object and r is the distance to it.

1. Calculate the angular area of the Sun as seen from the Earth. [1]

   We also define the **surface brightness** of an object as $\Sigma = f/\Omega$, where f is the flux received from it.

2. Calculate the flux received and the surface brightness of the Sun at Earth. [2]

   Consider a static, infinite, eternal Euclidean universe uniformly filled with Sun-like stars. Each star has radius $R_\star = 7.0 \times 10^8\,\mathrm{m}$, luminosity $L_\star = 3.83 \times 10^{26}\,\mathrm{W}$, and a constant stellar number density $n = 2 \times 10^8\,\mathrm{Mpc}^{-3}$.

3. Consider a spherical shell of radius r and thickness $\Delta r = 1\,\mathrm{Mpc}$ centred on Earth. Calculate the number of stars in this shell at $r = 1000\,\mathrm{Mpc}$. [1]

4. Calculate the flux received at Earth from all stars in this shell, showing that it is independent of r. What would be the total flux received at Earth in an infinite universe? [3]

5. For what distance r_{Olb} in Mpc (i.e. how many shells) would the sky be completely paved by stars? Calculate the surface brightness of the sky if the size of the universe were r_{Olb}. [4]

   If the universe extends to $r \geqslant r_{\mathrm{Olb}}$, the sky must be uniformly bright with a surface brightness equal to that of a typical star. Obviously it is not; at least one assumption must be wrong. One possibility is that the universe is not infinite in size.

   The **Hubble–Lemaître Law** states that a galaxy at distance d recedes from us at speed $v = H_0\,d$, where H_0 is the Hubble constant. The **Hubble radius** d_H marks the boundary beyond which the recession speed exceeds the speed of light; this provides a good approximation for the size of the observable universe.

6. Ignoring cosmological redshift, compute the average surface brightness of the sky for a universe of size d_H. [3]

Solutions

Solution a

The Sun is a disc of radius $R_\odot$ at distance $a_\oplus$, so its solid angle is

$$\Omega_\odot = \frac{\pi R_\odot^2}{a_\oplus^2} = \frac{\pi\,(6.955\times10^8\ \mathrm{m})^2}{(1.496\times10^{11}\ \mathrm{m})^2} \approx \boxed{6.79\times10^{-5}\ \mathrm{sr}}$$

[Correct formula and substitution **1 Mark**]

Solution b

Flux:

$$f_\odot = \frac{L_\odot}{4\pi a_\oplus^2} = \frac{3.826\times10^{26}\ \mathrm{W}}{4\pi\,(1.496\times10^{11}\ \mathrm{m})^2} \approx \boxed{1361\ \mathrm{W\,m^{-2}}}$$

[Correct formula and numerical value **1 Mark**]

Surface brightness:

$$\Sigma_\odot = \frac{f_\odot}{\Omega_\odot} = \frac{1361\ \mathrm{W\,m^{-2}}}{6.79\times10^{-5}\ \mathrm{sr}} \approx \boxed{2.00\times10^{7}\ \mathrm{W\,m^{-2}\,sr^{-1}}}$$

[Correct formula and numerical value **1 Mark**]

Solution c

Volume of the shell: $V_{\mathrm{shell}} = 4\pi r^2\,\Delta r$. With $r = 1000\,\mathrm{Mpc}$ and $\Delta r = 1\,\mathrm{Mpc}$,

$$N = n\times4\pi r^2\,\Delta r = 2\times10^8\ \mathrm{Mpc^{-3}}\times4\pi\times(10^3\ \mathrm{Mpc})^2\times1\ \mathrm{Mpc} \approx \boxed{2.51\times10^{15}\ \mathrm{stars}}$$

[Correct expression **0.5 Mark**]; [Correct numerical value **0.5 Mark**]

Solution d

Flux at Earth from a single star at distance r:

$$f_\star(r) = \frac{L_\star}{4\pi r^2}.$$

Number of stars in the shell: $N = 4\pi r^2\,n\,\Delta r$. Total flux from the shell:

$$\Delta f = N\cdot f_\star(r) = 4\pi r^2\,n\,\Delta r\times\frac{L_\star}{4\pi r^2} = n\,L_\star\,\Delta r.$$

[Setting up the product $N\cdot f_\star$ **1 Mark**]

The r^2 factors cancel exactly; Δf is **independent of** r—every shell contributes equally regardless of distance. [Showing cancellation and conclusion **1 Mark**]

Numerically (using $1\,\mathrm{Mpc} = 3.086\times10^{22}\,\mathrm{m}$):

$$n_{\mathrm{SI}} = \frac{2\times10^8}{(3.086\times10^{22})^3} = 6.81\times10^{-60}\ \mathrm{m^{-3}}, \qquad \Delta f = 6.81\times10^{-60}\times3.83\times10^{26}\times3.086\times10^{22} \approx 8.05\times10^{-11}\ \mathrm{W\,m^{-2}}$$

per 1 Mpc shell. Since each shell contributes the same $\Delta f > 0$ and there are infinitely many shells, the total flux from an infinite universe diverges:

$$f_{\mathrm{total}} = \int_0^\infty n\,L_\star\,dr \longrightarrow \infty.$$

[Conclusion of divergence in an infinite universe **1 Mark**]

Solution e

The sky is completely paved when the total solid angle subtended by all stars equals 4π sr. The solid angle of one star at distance r is $\pi R_\star^2/r^2$, so integrating over all shells:

$$\int_0^{r_{\rm Olb}} n \cdot 4\pi r^2 \cdot \frac{\pi R_\star^2}{r^2}\, dr = 4\pi^2\, n\, R_\star^2\, r_{\rm Olb} = 4\pi.$$

[Setting up the solid-angle condition **1 Mark**]

Solving for $r_{\rm Olb}$:

$$\boxed{r_{\rm Olb} = \frac{1}{\pi\, n\, R_\star^2}}$$

[Correct expression **1 Mark**]

Substituting SI values:

$$r_{\rm Olb} = \frac{1}{\pi \times 6.81 \times 10^{-60}\ \mathrm{m^{-3}} \times (7.0 \times 10^8\ \mathrm{m})^2} = \frac{1}{\pi \times 3.34 \times 10^{-42}\ \mathrm{m^{-1}}} \approx 9.54{\times}10^{40}\ \mathrm{m} \approx \boxed{3.09 \times 10^{18}\ \mathrm{Mpc}}.$$

[Correct numerical value **1 Mark**]

At $r = r_{\rm Olb}$ every line of sight terminates on a stellar surface, so the sky surface brightness equals the intrinsic surface brightness of a star:

$$\Sigma_{\rm sky} = \Sigma_\star = \frac{f_\star}{\Omega_\star} = \frac{L_\star/(4\pi r^2)}{\pi R_\star^2/r^2} = \frac{L_\star}{4\pi^2 R_\star^2} = \frac{3.83 \times 10^{26}}{4\pi^2 \times (7.0 \times 10^8)^2} \approx \boxed{1.98 \times 10^7\ \mathrm{W\,m^{-2}\,sr^{-1}}.}$$

[Correct expression and value **1 Mark**]

This equals $\Sigma_\odot$ from part (b)—as expected, since Sun-like stars were used.

Solution f

Hubble radius. Using $H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$:

$$d_H = \frac{c}{H_0} = \frac{3.0 \times 10^5\ \mathrm{km\,s^{-1}}}{70\ \mathrm{km\,s^{-1}\,Mpc^{-1}}} \approx 4286\ \mathrm{Mpc} = 1.32 \times 10^{26}\ \mathrm{m}.$$

[Correct formula and Hubble radius **1 Mark**]

Total flux. Using the result from part (d):

$$f = \int_0^{d_H} n\, L_\star\, dr = n\, L_\star\, d_H = 6.81 \times 10^{-60} \times 3.83 \times 10^{26} \times 1.32 \times 10^{26} \approx 3.44 \times 10^{-7}\ \mathrm{W\,m^{-2}}.$$

[Correct integral and numerical value **1 Mark**]

Average surface brightness. If the sky has uniform surface brightness Σ, then $f = \Sigma \times 4\pi$, so:

$$\Sigma_{\rm sky} = \frac{f}{4\pi} = \frac{3.44 \times 10^{-7}}{4\pi} \approx \boxed{2.74 \times 10^{-8}\ \mathrm{W\,m^{-2}\,sr^{-1}}.}$$

[Correct formula and value **1 Mark**]

This is $\sim 10^{15}$ times fainter than the Sun's surface brightness from part (b), consistent with the dark night sky we actually observe.
[Accept any final value in the range 10^{-8}–$10^{-7}\,\mathrm{W\,m^{-2}\,sr^{-1}}$ with correct method.]

3 Astrometric Imaging and Binary Inference [19]

On a clear day, the astronomer Tonmay uses a telescope of focal ratio $f/3$, aperture 150 cm, and a CCD plate of size (1.5×1.5) cm. Each pixel of the CCD is a square of side $l = 1\ \mu$m. The focal length of the telescope is therefore $f = 450$ cm.
Assume that the angular separations involved are sufficiently small that right ascension differences may be treated as linear angular separations.
First, he observes a compact three-star system, MEDAL, whose components have the following equatorial coordinates:

Table 1: Equatorial coordinates of the three stars in the MEDAL system.

Star	Right Ascension	Declination
PM	$\alpha_1 = 18^{\mathrm{h}}56^{\mathrm{m}}35.40^{\mathrm{s}}$	$\delta_1 = 35°27'32.50''$
DY	$\alpha_2 = 18^{\mathrm{h}}56^{\mathrm{m}}33.50^{\mathrm{s}}$	$\delta_2 = 35°27'32.15''$
TY	$\alpha_3 = 18^{\mathrm{h}}56^{\mathrm{m}}37.15^{\mathrm{s}}$	$\delta_3 = 35°27'33.30''$

1. On a plot of declination versus right ascension, represent the arrangement of the three stars in the MEDAL system. At which coordinates should Tonmay point his telescope to best centre the system? [3]
2. Assuming that the region enclosed by the three stars is fully sampled on the CCD, determine the total number of pixels occupied by this region. [4]

Later, using the same telescope and CCD, Tonmay observes a different field containing four stars at the same distance of 30.1 pc from Earth. At the first epoch, the stars have the angular positions listed in Table 2. Fifty years later, the same region is observed again by Tonmay's grandchild, Sajid, in the same fixed coordinate system, and only three distinct angular positions are detected, as listed in Table 3; one of the stars is no longer resolved as a separate source.

Table 2: Observed angular positions at the first epoch.

Star	x (arcsec)	y (arcsec)
A	-0.50	$+0.50$
B	$+0.20$	$+0.50$
C	-0.492	-0.50
D	$+0.492$	-0.50

Table 3: Distinct observed angular positions at the second epoch.

Point	x (arcsec)	y (arcsec)
P1	-0.15	$+0.85$
P2	-0.15	$+0.15$
P3	0.00	-0.50

Assume that the four stars form two independent binary systems and that, apart from their orbital motion within these binaries, the stars have no proper motion. The centre of mass of each binary remains fixed in the sky coordinate system. Assume further that all orbits are circular, that the configuration in the first epoch corresponds to the maximum projected separation in both binaries, and that for each binary the normal vector to the orbital plane is either parallel to the line of sight or parallel to the vertical axis in the plane of the sky.

3. Using the first-epoch coordinates, determine the pairing of the four stars into the two binary systems. [3]

4. Using the 50-year time interval and the second-epoch coordinates, determine the possible orbital periods of the two binary systems as functions of an integer parameter, or show that they cannot be uniquely determined. [4]

5. Using the distance 30.1 pc and Kepler's third law, determine the stellar masses M_A, M_B, M_C, and M_D in terms of the same integer parameter, or show that they cannot be uniquely determined. [5]

Solutions

Solution a

The best pointing direction minimises the maximum angular offset to any member of the system, which for an unweighted triangular arrangement is the centroid. Average the right ascensions and declinations separately.

Right ascension:

$$\bar{\alpha} = \frac{18^{\mathrm{h}}56^{\mathrm{m}}35.40^{\mathrm{s}} + 18^{\mathrm{h}}56^{\mathrm{m}}33.50^{\mathrm{s}} + 18^{\mathrm{h}}56^{\mathrm{m}}37.15^{\mathrm{s}}}{3}.$$

Summing the seconds: $35.40 + 33.50 + 37.15 = 106.05$, giving

$$\bar{\alpha} = 18^{\mathrm{h}}56^{\mathrm{m}}35.35^{\mathrm{s}}.$$

Declination:

$$\bar{\delta} = \frac{35^\circ 27' 32.50'' + 35^\circ 27' 32.15'' + 35^\circ 27' 33.30''}{3}.$$

Summing the arcseconds: $32.50 + 32.15 + 33.30 = 97.95$, giving

$$\bar{\delta} = 35^\circ 27' 32.65''.$$

Tonmay should point his telescope at

$$\boxed{(\bar{\alpha},\, \bar{\delta}) = \left(18^{\mathrm{h}}56^{\mathrm{m}}35.35^{\mathrm{s}},\ 35^\circ 27' 32.65''\right).}$$

[Correct method (centroid) **1 Mark**]; [Correct $\bar{\alpha}$ **1 Mark**]; [Correct $\bar{\delta}$ **1 Mark**]

Solution b

Plate scale. The focal length is $f = 450$ cm and the pixel side is $l = 1\ \mu$m $= 10^{-4}$ cm. The angular size of one pixel is

$$\theta_{\mathrm{pix}} = \frac{l}{f} = \frac{10^{-4}}{450}\ \mathrm{rad} = \frac{10^{-4}}{450} \times 206265 \approx 0.04584''.$$

Angular coordinates of the triangle. Taking DY as origin and converting RA differences to arcseconds (multiply seconds of time by 15):

$$(\Delta x,\, \Delta y)_{\mathrm{PM}} = \big((35.40 - 33.50) \times 15,\ 32.50 - 32.15\big) = (28.50'',\ 0.35''),$$

$$(\Delta x,\, \Delta y)_{\mathrm{TY}} = \big((37.15 - 33.50) \times 15,\ 33.30 - 32.15\big) = (54.75'',\ 1.15'').$$

Area of the triangle. Using the cross-product formula:

$$A = \tfrac{1}{2}\,|x_1 y_2 - y_1 x_2| = \tfrac{1}{2}\,|28.50 \times 1.15 - 0.35 \times 54.75| = \tfrac{1}{2}\,|32.775 - 19.163| = 6.806\ \mathrm{arcsec}^2.$$

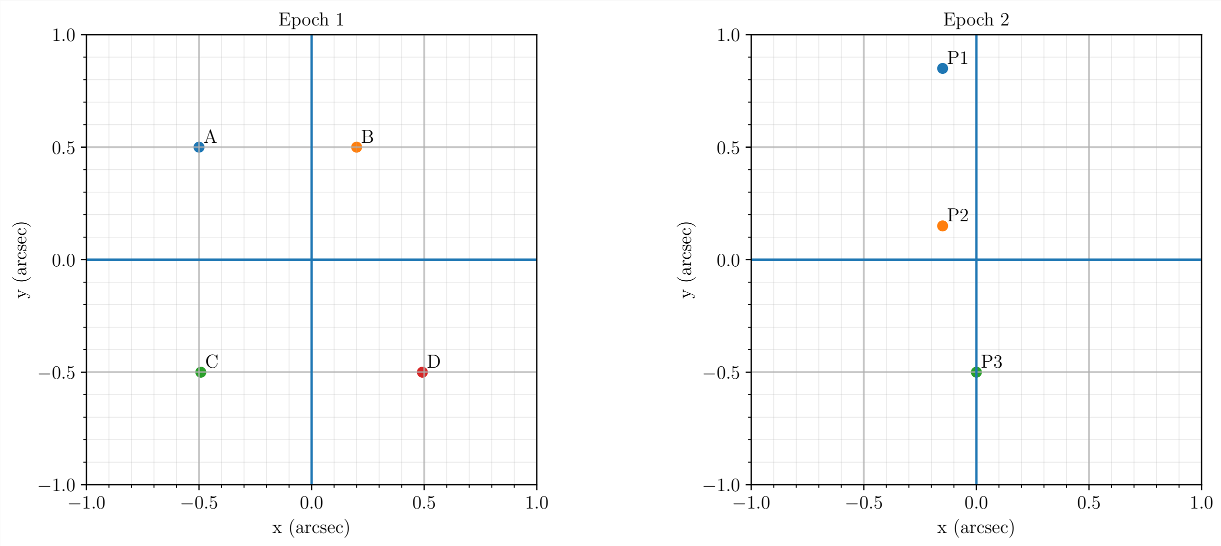
[Correct plate scale **1 Mark**]; [Correct angular coordinates **1 Mark**]; [Correct area **1 Mark**]

Number of pixels.

$$N = \frac{A}{\theta_{\text{pix}}^2} = \frac{6.806}{(0.04584)^2} \approx \boxed{3.24 \times 10^3 \text{ pixels.}}$$

[Correct pixel count **1 Mark**]

Solution c



Epcoh 1 & Epoch 2

Pairwise separations at the first epoch:

$$d_{AB} = \sqrt{(0.70)^2 + (0.00)^2} = 0.70'',$$

$$d_{CD} = \sqrt{(0.984)^2 + (0.00)^2} = 0.984'',$$

$$d_{AC} = \sqrt{(-0.008)^2 + (1.00)^2} \approx 1.000'', \quad d_{BD} = \sqrt{(0.292)^2 + (1.00)^2} \approx 1.042'',$$

$$d_{BC} = \sqrt{(0.692)^2 + (1.00)^2} \approx 1.215'', \quad d_{AD} = \sqrt{(0.992)^2 + (1.00)^2} \approx 1.409''.$$

The pairs (A, B) and (C, D) have the two smallest separations and share a common y-coordinate within each pair, identifying them as the natural binaries.

[Computing at least four pairwise distances **1 Mark**]

This identification is confirmed by the second epoch. The midpoint of P1 and P2 is

$$\left(\frac{-0.15 - 0.15}{2}, \frac{0.85 + 0.15}{2}\right) = (-0.15,\ 0.50),$$

which equals the first-epoch midpoint of A and B:

$$\left(\frac{-0.50 + 0.20}{2}, \frac{0.50 + 0.50}{2}\right) = (-0.15,\ 0.50).$$

Similarly, P3 $= (0.00,\ -0.50)$ equals the first-epoch midpoint of C and D:

$$\left(\frac{-0.492 + 0.492}{2}, \frac{-0.50 - 0.50}{2}\right) = (0.00,\ -0.50).$$

$$\boxed{\text{The two binaries are } (A, B) \text{ and } (C, D).}$$

[Correct pairing by separation argument **1 Mark**]; [Confirmed by centre-of-mass check **1 Mark**]

Solution d

For the binary (A, B), the first epoch shows the maximum projected separation along the horizontal axis; at the second epoch the two stars appear at P1 and P2, which are separated along the *vertical* axis. The relative position vector has therefore rotated by an odd multiple of 90° in 50 years:

$$2\pi \frac{50}{P_{AB}} = \frac{(2n_{AB}+1)\pi}{2}, \qquad n_{AB} = 0, 1, 2, \ldots \implies \boxed{P_{AB} = \frac{200}{2n_{AB}+1}\ \text{yr.}}$$

For the binary (C, D), the first epoch again shows maximum projected separation, but at the second epoch the pair is unresolved (both stars merge into P3). The projected separation has collapsed to zero, which also requires an odd quarter-turn advance:

$$2\pi \frac{50}{P_{CD}} = \frac{(2n_{CD}+1)\pi}{2}, \qquad n_{CD} = 0, 1, 2, \ldots \implies \boxed{P_{CD} = \frac{200}{2n_{CD}+1}\ \text{yr.}}$$

The periods cannot be uniquely determined; they form one-parameter families indexed by the non-negative integers n_{AB} and n_{CD} independently.

[Correct quarter-turn argument for (A, B) **2 Mark**]; [Correct result for (C, D) **2 Mark**]

Solution e

For a binary of total mass M_{tot}, semimajor axis a (in AU), and period P (in years), Kepler's third law gives

$$M_{\text{tot}} = \frac{a^3}{P^2}\, M_\odot.$$

Since each binary's centre of mass is fixed and the problem states equal projected separations about the midpoint, the two stars in each pair are equidistant from the centre of mass, hence $M_1 = M_2$.

Binary (A, B). The maximum projected separation is $d_{AB} = 0.70''$. Converting to physical separation at 30.1 pc:

$$a_{AB} = 0.70 \times 30.1 = 21.07\ \text{AU}.$$

With $M_A = M_B$:

$$2M_A = \frac{(21.07)^3}{P_{AB}^2}\, M_\odot = \frac{(21.07)^3}{\left(200/(2n_{AB}+1)\right)^2}\, M_\odot,$$

$$\boxed{M_A = M_B = \frac{(21.07)^3}{80000}(2n_{AB}+1)^2\ M_\odot \approx 0.1169\,(2n_{AB}+1)^2\ M_\odot.}$$

[Correct conversion and Kepler application for (A, B) **2 Mark**]

Binary (C, D). The maximum projected separation is $d_{CD} = 0.984''$:

$$a_{CD} = 0.984 \times 30.1 = 29.62\ \text{AU}.$$

With $M_C = M_D$:

$$2M_C = \frac{(29.62)^3}{\left(200/(2n_{CD}+1)\right)^2}\, M_\odot,$$

$$\boxed{M_C = M_D = \frac{(29.62)^3}{80000}(2n_{CD}+1)^2\ M_\odot \approx 0.3248\,(2n_{CD}+1)^2\ M_\odot.}$$

[Correct conversion and Kepler application for (C, D) **2 Mark**]; [Correct conclusion that masses cannot be uniquely determined **1 Mark**]

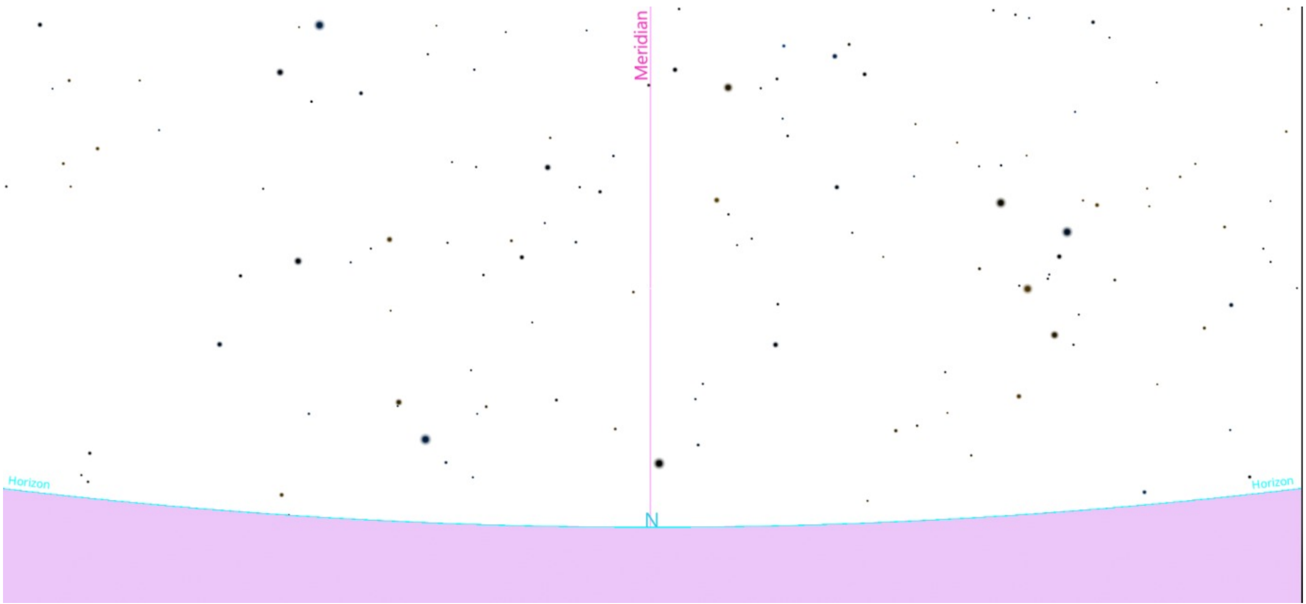
The stellar masses are not uniquely determined; for the physically most plausible case $n_{AB} = n_{CD} = 0$ one obtains $M_A = M_B \approx 0.12\, M_\odot$ and $M_C = M_D \approx 0.32\, M_\odot$.

The Lost Years Written in the Stars [30 marks]

On 3 June 2019, the world as we know it suddenly froze. Every human was encased in stone, their lives halted in an instant. Centuries passed. Forests reclaimed cities. Rivers carved new paths. Stars continued their eternal dance across the sky, unbothered by humanity's fate.
Nearly three millennia later, one human finally awakens: **Senku Ishigami**, a boy of science. Alone under the vast night sky, he knows one thing—to rebuild civilisation, he must first rebuild knowledge. But the heavens he sees are no longer quite the same. The *Pole Star has drifted*, constellations are slightly twisted, and even the cardinal points have shifted with the Earth's slow precession.
Astonishingly, Senku kept track of the time during the thousands of years in which he was petrified so that he could wake up in an appropriate season for survival. Now he wants to verify the current date using the stars. Senku recalls: *"The stars never lie. If I can measure how far the North Celestial Pole has shifted, I can calculate how long I've slept…and what year this is."*
Because of Earth's axial **precession**, the NCP drifts around the ecliptic pole with a period of $\sim$ 26,000 years. Proper motions of stars may be neglected.

Part (A): Understanding Precession [8 marks]

Before you step in to help Senku, you must have an idea of how the celestial pole drifts over time. In Part (A), you will rediscover these concepts by working through a few subtasks.
For this part, consider the given horizon-view sky map of a certain region of the sky close to the local horizon from the epoch 2019, i.e. before the petrification. Assume that the curvature of the celestial sphere is negligible in the image, so there are no distortions in the presented map.



1. Locate the position of the star Thuban (α Dra) in the image. [1]

2. Knowing that the North Celestial Pole in the past had practically the same position as Thuban, whose declination is $\delta_{\text{Thuban}} = 64°16.5'$, and knowing the obliquity of the ecliptic $\epsilon = 23°27'$, find the position of the North Ecliptic Pole. [2.5]

3. In 5400 years, which star will be closest to the North Celestial Pole? Calculate the angular separation between these points; this will correspond to the angular error of the pole after 5400 years. Given: angular velocity of the vernal point relative to the stars $\omega_\gamma = 50.2$ arcsec/year. [2]

4. Knowing that the right ascension of Thuban is $\alpha_T = 14^{\text{h}}5^{\text{m}}$ and that the right ascension of the Sun at the time of the image is $\alpha_\odot = 4^{\text{h}}32.8^{\text{m}}$, calculate the time at which the image was taken, i.e. the local apparent solar time. [2.5]

Part (B): Rediscovering the Date and Year [22 marks]

The sky map made by Senku on the night he woke up is shown below. The latitude of the observing site is 45° N as estimated from the sky map.

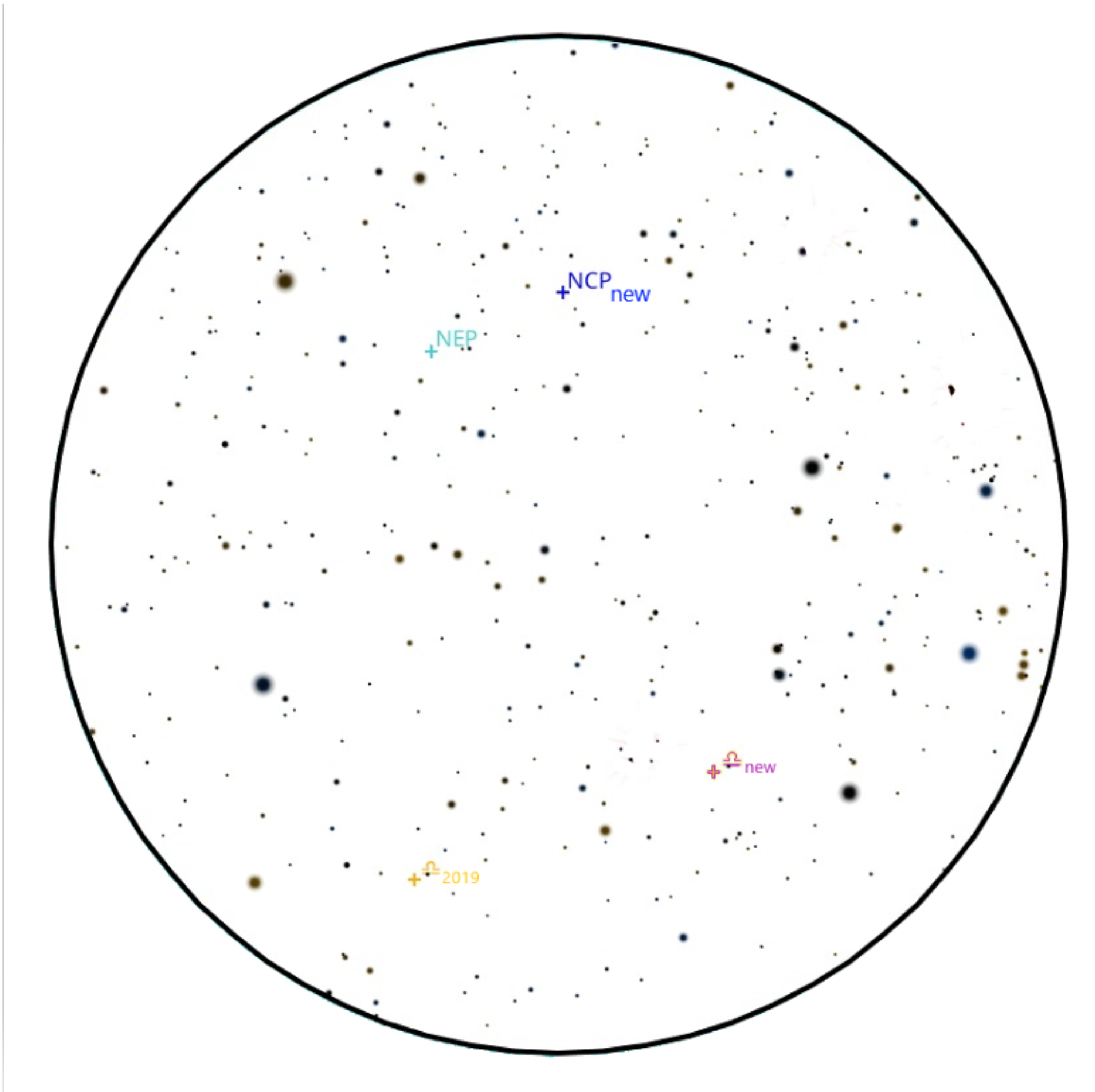


Figure 1: Sky map corresponding to the night Senku woke up.

5. Since Senku cannot rely on the celestial equator of epoch 2019, he attempts to trace the unchanged **ecliptic** on the sky, on which the equinoxes have shifted from their original positions. Now: (i) draw the ecliptic, and (ii) state in which constellation the new autumnal equinox resides (in 2019 it was in Virgo). [3]

   By observing that the autumnal equinox is visible at midnight, Senku infers that the Sun must lie in the opposite direction, i.e. near the vernal equinox. He now aims to determine the Local Sidereal Time, which can be obtained from the hour angle of the equinox. However, since the celestial equator is no longer reliable, he adopts an alternative approach.

6. Estimate the altitude and azimuth of both the new and old (2019) autumnal equinox, and record the values in the table below. [4]

	Altitude	Azimuth
Old equinox		
New equinox		

7. Using these values, determine the hour angle of the vernal equinox, i.e. the **Local Sidereal Time** at the moment the sky map was taken. [6]
8. Senku attempts the key calculation: *"If I can measure the pole shift, I can measure the years I have lost."* Determine the **approximate year** in which Senku awakened. [6]
9. Using the local sidereal time, estimate the present date, assuming the observation was made at local midnight. If needed, take $\phi = 45°$ N. [3]

Solutions

Part (A)

Solution f: question

Below we see the star Thuban marked:

[Correct identification **1 mark**]

Solution g: question

The North Celestial Pole practically passed through the positions of Polaris and Thuban, so both stars lie on the precession circle whose centre is the North Ecliptic Pole. Since we may neglect the curvature of the celestial sphere, all angular distances are proportional to linear distances in the image.
The angular distance from Thuban to the North Celestial Pole is

$$\Delta_T = 90° - 64°16.5' = 25°43.3'.$$

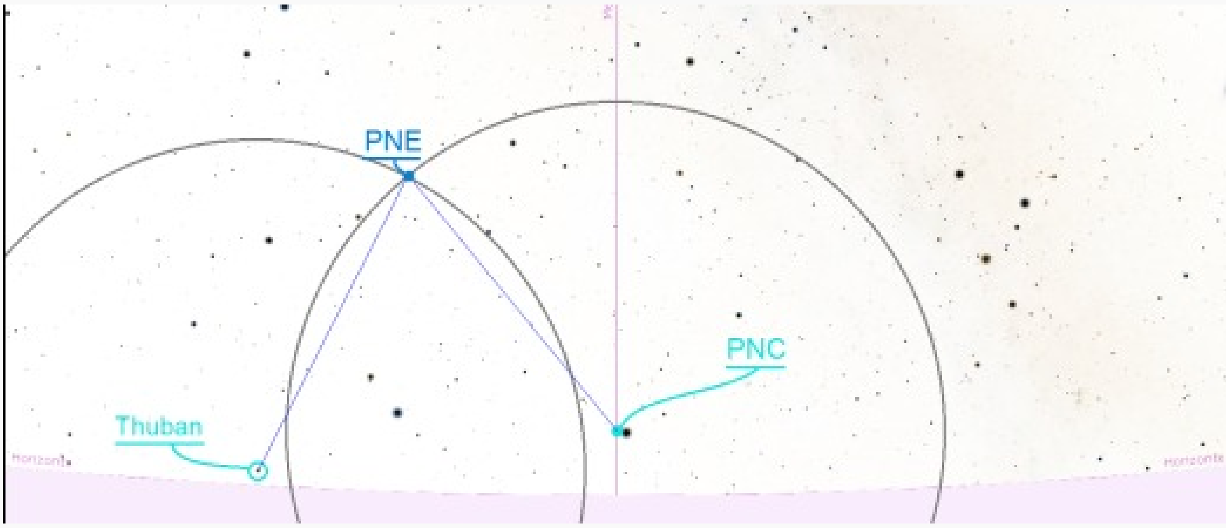
[Correct Δ_T **0.5 mark**]

The angular distance between the North Ecliptic Pole and the North Celestial Pole equals the obliquity $\epsilon = 23°27'$. Using pixel (or ruler) measurements in the image, if the Thuban–NCP distance spans d_{TC} (pixels), then the NEP–NCP distance is

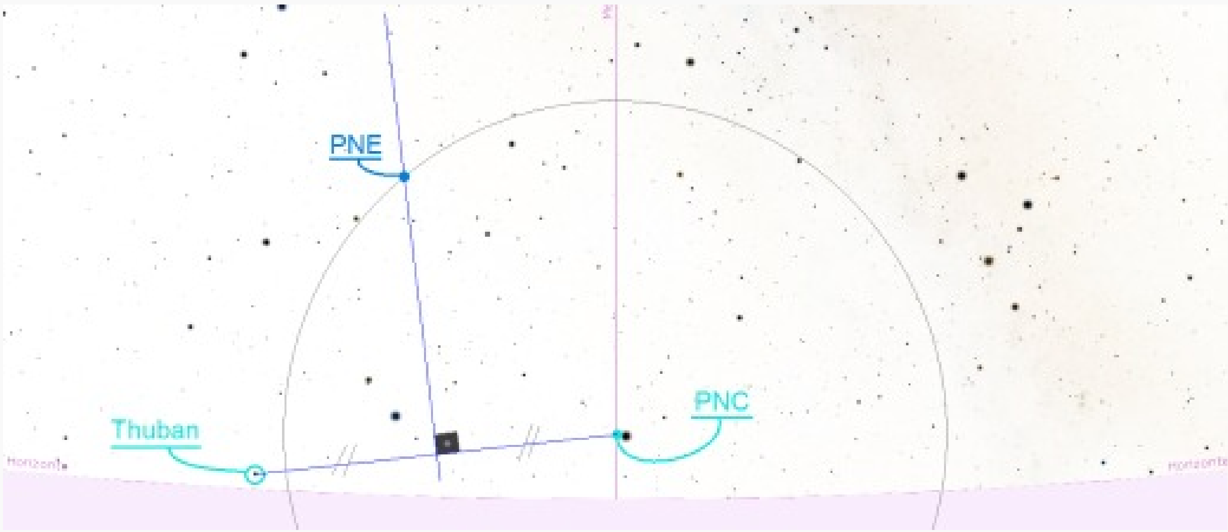
$$d_{EC} = d_{TC} \times \frac{23°27'}{25°43.3'} \approx 0.9117\, d_{TC}.$$

[Correct ratio $d_{EC}/d_{TC} \approx 0.912$, regardless of units **1 mark**]

Drawing two arcs of radius d_{EC} centred on Polaris and Thuban respectively, their intersection gives the North Ecliptic Pole:



[Correct location of NEP marked in image **1 mark**]



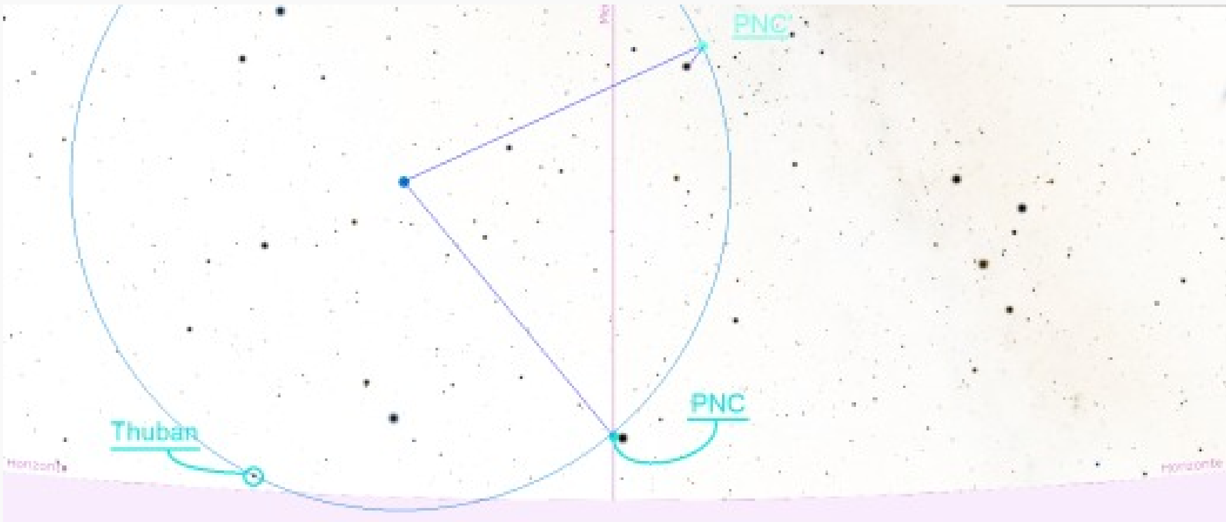
Solution h: question

In 5400 years the NCP sweeps an angle θ along the precession circle:

$$\theta = \frac{5400\,\text{yr} \times 50.2\ \text{arcsec/yr}}{3600\ \text{arcsec/deg}} = 75.3^\circ.$$

[Correct precession angle **0.5 mark**]

Shifting the current NCP (Polaris) by 75.3° around the NEP gives the future pole position NCP′, shown below. The star closest to NCP′ is **Alderamin** (α Cep). [Correct identification of Alderamin **1 mark**]



Using proportionality with d_{TC} as reference (linear distance Thuban to NCP in image), the angular error between Alderamin and NCP′ is

$$\xi = \frac{d_{\text{star}-\text{NCP}'}}{d_{TC}} \times \Delta_T \approx \frac{30.61}{403.41} \times 25^\circ 43.3' \approx 1^\circ 57'.$$

(Any consistent pixel or ruler scale gives the same ratio; the key figure is $d_{\text{star}-\text{NCP}'}/d_{TC} \approx 0.0759$.) [Correct angular error **0.5 mark**]

Solution i: question

The sidereal time equation relates the hour angles and right ascensions of any two objects:

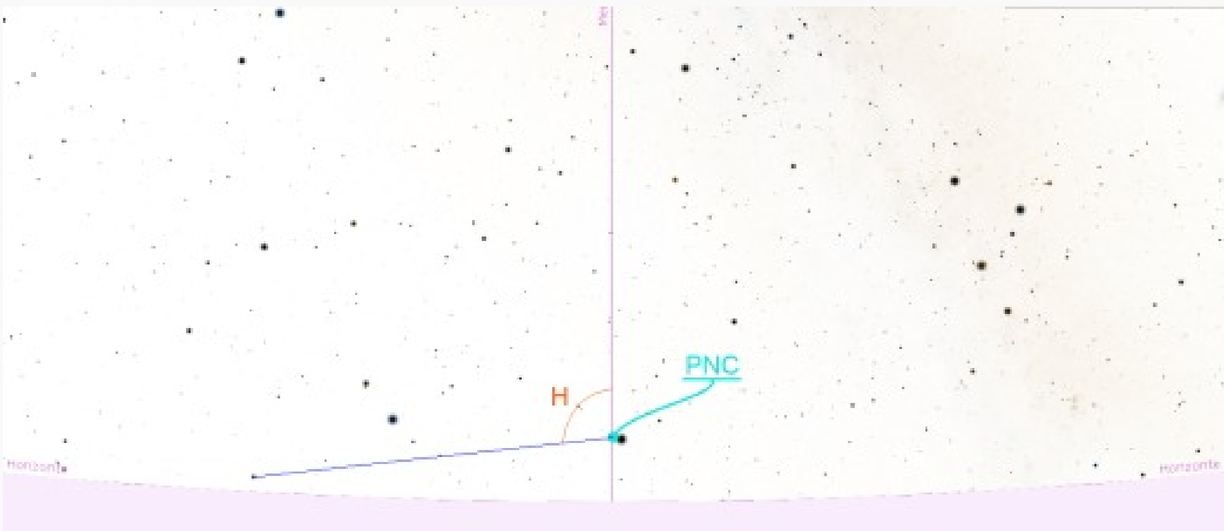
$$\text{LST} = H_\star + \alpha_\star.$$

[Correct relation stated **0.5 mark**]

Applying it to both Thuban and the Sun:

$$H_\odot + \alpha_\odot = H_T + \alpha_T.$$

From the image, the hour angle of Thuban is read off as $H_T \approx 96.7^\circ = 6^\text{h}26.8^\text{m}$. [Correct reading of H_T from image **1 mark**]



Substituting:

$$H_\odot = H_T + \alpha_T - \alpha_\odot = 6^\text{h}26.8^\text{m} + 14^\text{h}5^\text{m} - 4^\text{h}32.8^\text{m} = 16^\text{h}.$$

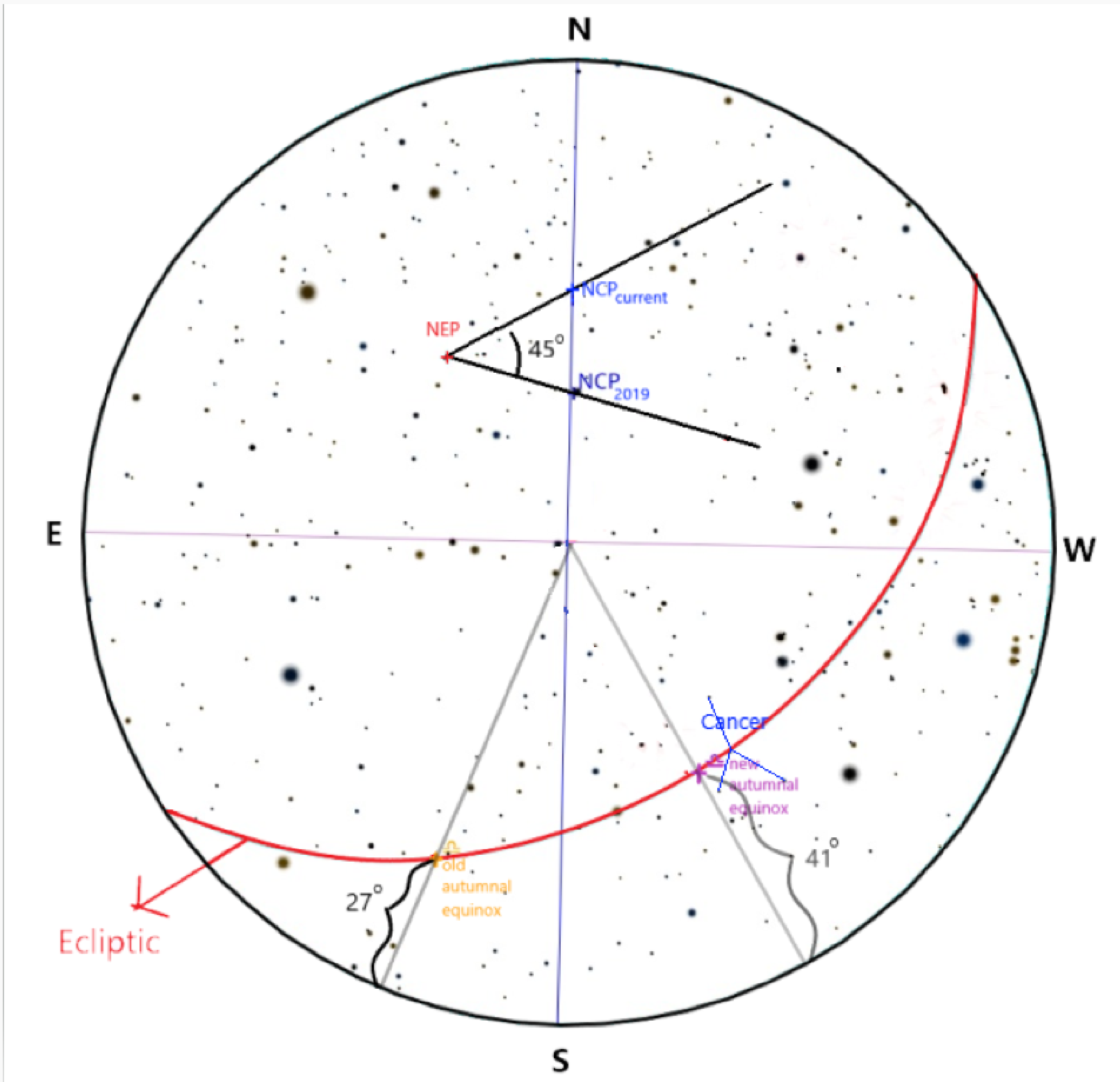
The local apparent solar time is $\mathrm{LAST} = H_{\odot} + 12^{\mathrm{h}}$, so

$$\mathrm{LAST} = 28^{\mathrm{h}} \equiv \boxed{4^{\mathrm{h}}.}$$

[Correct final solar time **1 mark**]

Part (B)

Solution j: question



The new autumnal equinox is located in the constellation of **Cancer**. [Correct constellation **1 mark**]

[Ecliptic drawn correctly **2 marks**]

Solution k: question

	Altitude	Azimuth
Old equinox	$27^\circ \pm 2^\circ$	$250^\circ \pm 3^\circ$
New equinox	$41^\circ \pm 2^\circ$	$301^\circ \pm 3^\circ$

[1 mark for each reading within the stated tolerance; total $4 \times 1 = 4$ marks]

Solution l: question

Given:

- Observer latitude: $\phi = 45^\circ$ N
- Altitude of the new autumnal equinox: $h = 41^\circ$
- Declination of the autumnal equinox: $\delta = 0^\circ$
- The autumnal equinox is **east** of the meridian (about to transit)

The altitude–hour-angle relation for an object on the celestial equator ($\delta = 0^\circ$):

$$\sin h = \sin\phi \sin\delta + \cos\phi \cos\delta \cos H = \cos\phi \cos H.$$

[Diagram of relevant celestial sphere **1 mark**]; [Correct relation written **1 mark**]

Solving for H:

$$\cos H = \frac{\sin h}{\cos\phi} = \frac{\sin 41^\circ}{\cos 45^\circ} \implies H \approx 21.9^\circ = 1^{\mathrm{h}}28^{\mathrm{m}}.$$

[Correct value of $|H|$ **2 marks**]

Since the equinox is east of the meridian (yet to transit):

$$H = -1^{\mathrm{h}}28^{\mathrm{m}}.$$

[Correct sign of H **1 mark**]

The Local Sidereal Time equals the RA of the vernal equinox plus its hour angle. The autumnal equinox has RA $= 12^{\mathrm{h}}$ (opposite the vernal equinox), so:

$$\mathrm{LST} = \mathrm{RA}_{\mathrm{aut.eq.}} + H = 12^{\mathrm{h}} + \left(-1^{\mathrm{h}}28^{\mathrm{m}}\right) = \boxed{10^{\mathrm{h}}32^{\mathrm{m}}.}$$

[Correct LST **1 mark**]

Solution m: question

Using a protractor on the sky map, the angle subtended at the North Ecliptic Pole between $\mathrm{NCP}_{\mathrm{new}}$ and NCP_{2019} is measured to be approximately 45°. [Correct angle read from map **1 mark**]

The number of years elapsed is

$$t = \frac{\text{angular drift}}{\text{precession rate}} = \frac{45^\circ}{360^\circ/26{,}000\ \mathrm{yr}} = \frac{45^\circ \times 26{,}000\ \mathrm{yr}}{360^\circ} \approx 3250\ \mathrm{yr}.$$

[Correct key relationship **1 mark**]

Adding to 2019:

$$\boxed{\text{Approximate wake-up year} \approx 2019 + 3250 = 5269\ \text{AD}.}$$

[Final answer within ± 100 years **4 marks**]

Solution n: question

- Observation time: local midnight (solar time $= 0^{\mathrm{h}}$)
- LST at this moment: $10^{\mathrm{h}}32^{\mathrm{m}}$

On the vernal equinox (approximately March 20), the LST at local midnight is $\approx 12^{\mathrm{h}}$ (the Sun is at RA $= 0^{\mathrm{h}}$, so the anti-solar direction has RA $= 12^{\mathrm{h}}$ on the meridian).
The difference:

$$\Delta\mathrm{LST} = 10^{\mathrm{h}}32^{\mathrm{m}} - 12^{\mathrm{h}} = -1^{\mathrm{h}}28^{\mathrm{m}}.$$

The negative sign indicates the current date is **before** the vernal equinox. [Correct sign interpretation **1 mark**]
Converting:

$$1^{\mathrm{h}}28^{\mathrm{m}} = 88 \text{ min}.$$

Each day the LST at midnight advances by ≈ 4 minutes, so

$$\text{Days before March 20} = \frac{88 \text{ min}}{4 \text{ min/day}} = 22 \text{ days}.$$

[Correct offset calculation **2 marks**]

$$\text{Date} = \text{March 20} - 22 \text{ days} = \boxed{\text{26 February.}}$$

[Final answer within ± 2 days **2 marks** (otherwise 1 mark)]
Note: in a leap year the result shifts to 27 February.